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# Laboratories — Department of Information Technology

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Source-type legend used in tables: **O** = official (quoted from a GUNI PDF), **DO** = derived_official (structured/transformed from a GUNI PDF), **S** = synthetic (invented for CampusMate testing). Official page numbers refer to physical PDF pages of DOC-OFFICIAL-SYLLABUS-001 (GUNI structured syllabus) or DOC-OFFICIAL-CATALOG-001 (GUNI 1.1.3 course list).

Ten fictional teaching laboratories of DEPT-IT-001. Buildings, rooms, equipment, hours and course associations are synthetic. Course codes/names shown are official.

## 1. Laboratory summary

source_type: **synthetic (CampusMate demo data)**

| Lab ID | Laboratory | Location | Capacity | Responsible faculty |
|---|---|---|---|---|
| LAB-IT-001 | AI & Machine Learning Lab | Demo Academic Block A, Floor 3, Room DA-301 | 36 | FAC-IT-002 Dr. Arjun Tensorkar |
| LAB-IT-002 | Natural Language Processing Lab | Demo Academic Block A, Floor 3, Room DA-302 | 30 | FAC-IT-003 Dr. Nisha Tokenwala |
| LAB-IT-003 | Database Systems Lab | Demo Academic Block A, Floor 3, Room DA-303 | 40 | FAC-IT-001 Dr. Meera Datavala |
| LAB-IT-004 | Programming Lab | Demo Academic Block A, Floor 3, Room DA-304 | 60 | FAC-IT-012 Prof. Ishaan Stackani |
| LAB-IT-005 | Networks & Security Lab | Demo Academic Block A, Floor 4, Room DA-401 | 32 | FAC-IT-007 Dr. Sana Cipherwala |
| LAB-IT-006 | Cloud & Big Data Lab | Demo Academic Block A, Floor 4, Room DA-402 | 36 | FAC-IT-009 Dr. Aditi Cloudani |
| LAB-IT-007 | IoT & Robotics Lab | Demo Innovation Block B, Floor 1, Room DB-101 | 24 | FAC-IT-016 Prof. Aman Circuitkar |
| LAB-IT-008 | Web & Mobile Development Lab | Demo Academic Block A, Floor 4, Room DA-403 | 40 | FAC-IT-015 Prof. Kavya Graphwala |
| LAB-IT-009 | Vision & Graphics Lab | Demo Academic Block A, Floor 4, Room DA-404 | 24 | FAC-IT-011 Prof. Tanya Pixelani |
| LAB-IT-010 | Systems & Software Engineering Lab | Demo Academic Block A, Floor 3, Room DA-305 | 40 | FAC-IT-014 Prof. Rohan Kernelwala |

## 2. Laboratory details

## LAB-IT-001 — AI & Machine Learning Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-001 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 3, Room DA-301 (ROOM-DA-301) |
| **capacity** | 36 students |
| **responsible faculty** | FAC-IT-002 Dr. Arjun Tensorkar |
| **opening hours** | Mon–Sat 09:00–18:00 (synthetic) |
| **equipment** | 36 workstations (32 GB RAM, RTX-class GPU); Python/Jupyter, PyTorch, scikit-learn; Shared model registry server |
| **related courses** | COURSE-2CEIT602 (ARTIFICIAL INTELLIGENCE); COURSE-2CEIT6PE4 (DATA SCIENCES AND MACHINE LEARNING); COURSE-2CEIT78PE1 (DEEP LEARNING); COURSE-3CEIT101 (COMPUTATIONAL INTELLIGENCE); COURSE-3CEIT309 (DEEP LEARNING) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-002 — Natural Language Processing Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-002 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 3, Room DA-302 (ROOM-DA-302) |
| **capacity** | 30 students |
| **responsible faculty** | FAC-IT-003 Dr. Nisha Tokenwala |
| **opening hours** | Mon–Fri 09:00–17:30 (synthetic) |
| **equipment** | 30 workstations; Local SLM inference server (8B-class models); Annotated corpora for Indic and English text; Vector database sandbox |
| **related courses** | COURSE-2CEIT78PE7 (NATURAL LANGUAGE PROCESSING); COURSE-3CEIT307 (NATURAL LANGUAGE PROCESSING); COURSE-3CEIT303 (SEMANTIC WEB AND SOCIAL NETWORKS); COURSE-2CEIT78PE5 (SEARCH ENGINE OPTIMIZATION) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-003 — Database Systems Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-003 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 3, Room DA-303 (ROOM-DA-303) |
| **capacity** | 40 students |
| **responsible faculty** | FAC-IT-001 Dr. Meera Datavala |
| **opening hours** | Mon–Sat 09:00–17:30 (synthetic) |
| **equipment** | 40 workstations; PostgreSQL and MySQL servers; MongoDB and Cassandra cluster; ER-modelling tools |
| **related courses** | COURSE-2CEIT303 (DATABASE MANAGEMENT SYSTEM); COURSE-2CEIT405 (NOSQL DATABASE SYSTEMS); COURSE-3CEIT211 (ADVANCED DATABASE MANAGEMENT SYSTEM); COURSE-3CEIT202 (DATA MINING AND DATA WAREHOUSING) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-004 — Programming Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-004 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 3, Room DA-304 (ROOM-DA-304) |
| **capacity** | 60 students |
| **responsible faculty** | FAC-IT-012 Prof. Ishaan Stackani |
| **opening hours** | Mon–Sat 08:30–19:00 (synthetic) |
| **equipment** | 60 workstations; JDK, .NET SDK, GCC, Python toolchains; Online judge server |
| **related courses** | COURSE-2CEIT302 (OBJECT ORIENTED PROGRAMMING); COURSE-2CEIT304 (DATA STRUCTURES); COURSE-2CEIT402 (DESIGN AND ANALYSIS OF ALGORITHMS); COURSE-2CEIT404 (PROGRAMMING WITH PYTHON); COURSE-2CEIT5PE1 (ADVANCED JAVA); COURSE-2CEIT5PE8 (ADVANCED PROGRAMMING WITH .NET); COURSE-2CEIT78PE9 (ADVANCED ALGORITHMS); COURSE-3CEIT102 (COMPUTER ALGORITHMS); COURSE-2CEIT5PE3 (QUANTUM COMPUTING); COURSE-3CEIT209 (ESSENTIAL MATHEMATICS); COURSE-3CEIT210 (MODELING AND SIMULATION); COURSE-3CEIT304 (DATA COMPRESSION) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-005 — Networks & Security Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-005 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 4, Room DA-401 (ROOM-DA-401) |
| **capacity** | 32 students |
| **responsible faculty** | FAC-IT-007 Dr. Sana Cipherwala |
| **opening hours** | Mon–Fri 09:00–17:00 (synthetic) |
| **equipment** | Managed switches and routers; Isolated penetration-testing network; Packet capture appliances; Wireless test bed |
| **related courses** | COURSE-2CEIT503 (COMPUTER NETWORKS); COURSE-2CEIT6PE3 (CRYPTOGRAPHY AND NETWORK SECURITY); COURSE-2CEIT6PE7 (ETHICAL HACKING); COURSE-2CEIT78PE8 (FORENSICS & CYBER LAW); COURSE-2CEIT78PE2 (BLOCKCHAIN TECHNOLOGIES); COURSE-3CEIT103 (CRYPTOGRAPHY AND NETWORK SECURITY); COURSE-3CEIT105 (WIRELESS NETWORKS); COURSE-3CEIT106 (ADVANCED TOPICS IN NETWORKS); COURSE-3CEIT107 (INTERNET TECHNOLOGIES); COURSE-3CEIT108 (NETWORK PROGRAMMING AND MANAGEMENT); COURSE-3CEIT207 (PUBLIC KEY INFRASTRUCTURE); COURSE-3CEIT310 (CYBER SECURITY); COURSE-3CEIT208 (SATELLITE NETWORKS) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-006 — Cloud & Big Data Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-006 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 4, Room DA-402 (ROOM-DA-402) |
| **capacity** | 36 students |
| **responsible faculty** | FAC-IT-009 Dr. Aditi Cloudani |
| **opening hours** | Mon–Sat 09:00–18:00 (synthetic) |
| **equipment** | Private OpenStack cloud; Hadoop/Spark cluster (8 nodes); Kubernetes sandbox |
| **related courses** | COURSE-2CEIT603 (CLOUD COMPUTING); COURSE-2CEIT702 (BIG DATA ANALYTICS); COURSE-2CEIT78PE10 (FOG COMPUTING); COURSE-2CEIT78PE11 (FAULT TOLERANT SYSTEMS); COURSE-2CEIT78PE12 (HIGH PERFORMANCE COMPUTING); COURSE-3CEIT109 (HIGH PERFORMANCE COMPUTING); COURSE-3CEIT110 (DISTRIBUTED COMPUTING); COURSE-3CEIT111 (SERVICE ORIENTED COMPUTING); COURSE-3CEIT203 (CLOUD COMPUTING); COURSE-3CEIT306 (BIG DATA ANALYTICS); COURSE-2CEIT6PE8 (IT INFRASTRUCTURE & MANAGEMENT) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-007 — IoT & Robotics Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-007 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Innovation Block B (BLDG-DEMO-B) |
| **floor / room** | Floor 1, Room DB-101 (ROOM-DB-101) |
| **capacity** | 24 students |
| **responsible faculty** | FAC-IT-016 Prof. Aman Circuitkar |
| **opening hours** | Mon–Fri 10:00–17:00 (synthetic) |
| **equipment** | Microcontroller and single-board computer kits; Sensor and actuator sets; Two mobile robot platforms; Digital electronics trainer kits |
| **related courses** | COURSE-2CEIT301 (DIGITAL ELECTRONICS); COURSE-2CEIT6PE6 (ROBOTICS); COURSE-2CEIT6PE9 (INTERNET OF THINGS); COURSE-3CEIT305 (INTERNET OF THINGS) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-008 — Web & Mobile Development Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-008 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 4, Room DA-403 (ROOM-DA-403) |
| **capacity** | 40 students |
| **responsible faculty** | FAC-IT-015 Prof. Kavya Graphwala |
| **opening hours** | Mon–Sat 09:00–18:00 (synthetic) |
| **equipment** | 40 workstations; Android and iOS device farm; Node.js/React toolchains; Usability testing station |
| **related courses** | COURSE-2CEIT403 (APPLICATION DEVELOPMENT TOOLS); COURSE-2CEIT5PE5 (MOBILE APPLICATION DEVELOPMENT); COURSE-2CEIT5PE6 (HUMAN COMPUTER INTERACTION); COURSE-2CEIT604 (USER EXPERIENCE DESIGN); COURSE-2CEIT6PE1 (WEB TECHNOLOGY); COURSE-2CEIT78PE4 (MOBILE COMPUTING); COURSE-2CEIT6PE5 (GAME DESIGN & DEVELOPMENT) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-009 — Vision & Graphics Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-009 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 4, Room DA-404 (ROOM-DA-404) |
| **capacity** | 24 students |
| **responsible faculty** | FAC-IT-011 Prof. Tanya Pixelani |
| **opening hours** | Mon–Fri 09:00–17:00 (synthetic) |
| **equipment** | GPU workstations; Stereo and depth cameras; VR headsets |
| **related courses** | COURSE-2CEIT5PE2 (COMPUTER GRAPHICS & VISUALIZATION); COURSE-2CEIT78PE3 (FUNDAMENTALS OF IMAGE PROCESSING); COURSE-2CEIT78PE6 (VIRTUAL REALITY); COURSE-2CEIT78PE13 (COMPUTER VISION); COURSE-3CEIT201 (DIGITAL IMAGE PROCESSING) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |

## LAB-IT-010 — Systems & Software Engineering Lab

source_type: **synthetic (CampusMate demo data)**

| | |
|---|---|
| **lab_id** | LAB-IT-010 |
| **department_id** | DEPT-IT-001 |
| **building** | Demo Academic Block A (BLDG-DEMO-A) |
| **floor / room** | Floor 3, Room DA-305 (ROOM-DA-305) |
| **capacity** | 40 students |
| **responsible faculty** | FAC-IT-014 Prof. Rohan Kernelwala |
| **opening hours** | Mon–Sat 09:00–17:30 (synthetic) |
| **equipment** | 40 Linux workstations; Virtualisation host; CI/CD and test-automation server; Compiler construction toolkits |
| **related courses** | COURSE-2CEIT401 (OPERATING SYSTEMS); COURSE-2CEIT501 (COMPUTER ARCHITECTURE & ORGANIZATION); COURSE-2CEIT502 (SOFTWARE ENGINEERING); COURSE-2CEIT601 (THEORY OF COMPUTATION); COURSE-2CEIT701 (COMPILER DESIGN); COURSE-2CEIT6PE2 (SOFTWARE VALIDATION & TESTING); COURSE-2CEIT5PE4 (SOFTWARE PACKAGES); COURSE-3CEIT112 (ADVANCE SOFTWARE ENGINEERING); COURSE-3CEIT113 (REAL TIME SYSTEMS); COURSE-3CEIT204 (ADVANCED OPERATING SYSTEM); COURSE-3CEIT205 (ADVANCED COMPUTER ARCHITECTURE); COURSE-3CEIT206 (SOFTWARE ARCHITECTURE AND DESIGN PATTERN); COURSE-3CEIT308 (COMPILER DESIGN) |
| **governing policies** | POLICY-002, POLICY-003 (Laboratory Usage Policy, Laboratory Safety Policy) |
| **booking contact** | it-labs@campusmate-demo.example |